Spectral–Spatial Masked Transformer With Supervised and Contrastive Learning for Hyperspectral Image Classification

Lingbo Huang, Yushi Chen, *Member, IEEE*, and Xin He

***Abstract*— Recently, due to the powerful capability at modeling the long-range relationships, Transformer-based methods have been widely explored in many research areas, including hyperspectral image (HSI) classification. However, because of lots of trainable parameters and the lack of inductive bias, it is difficult to train a Transformer-based HSI classifier, especially when the number of training samples is limited. To address this issue, in this study, spectral–spatial masked Transformer (SS-MTr) is explored for HSI classification, which uses a two-stage training strategy. In the first stage, SS-MTr pretrains a vanilla Transformer via reconstruction from masked HSI inputs, which embeds the local inductive bias into the Transformer. In the second stage, the well pretrained Transformer is cooperated with a fully connected layer and is then fine-tuned for the HSI classification. Furthermore, in order to incorporate discriminative feature learning into the SS-MTr, three SS-MTr-based methods, including contrastive SS-MTr (C-SS-MTr), supervised SS-MTr (S-SS-MTr), and supervised C-SS-MTr (SC-SS-MTr), are proposed by adding extra branches for specific tasks in parallel with the existing reconstruction task. Specifically, the proposed C-SS-MTr adds a contrastive loss, which brings instance discriminability. Besides, the proposed S-SS-MTr builds an extra classification branch for embracing interclass discriminability and intraclass similarity. Moreover, the proposed SC-SS-MTr combines C-SS-MTr and S-SS-MTr for better generalization. The proposed SS-MTr, C-SS-MTr, S-SS-MTr, and SC-SS-MTr are tested on three popular hyperspectral datasets (i.e., Indian Pines, Pavia University, and Houston). The obtained results reveal that the proposed models achieve competitive results compared with the state-of-the-art HSI classification methods. The code is available at https://github.com/mengduanjinghua/SS-MTr.**

***Index Terms*— Contrastive learning (CL), deep learning, hyperspectral image (HSI) classification, pretraining, Transformer.**

I. Introduction

HYPERSPECTRAL sensor uses imaging spectroscopy technology to capture abundant spectral and spatial information of the observing targets simultaneously. Therefore, the obtained hyperspectral image (HSI) is a useful tool in many real applications [1]. To make full use of HSI, data processing techniques, such as classification, unmixing, and detection, are vibrant topics in the field of remote sensing [2].

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HSI classification is a process of identifying each pixel in the scene into a predetermined category [3]. In the view of machine learning, classification tries to train a model with the help of training samples and then assign the input with a class label. Due to the fact that it is a fundamental technique of HSI processing, HSI classification is a hot topic in the remote sensing community for many years [4], [5].

In the early stage of HSI classification, most methods focused on handcrafted feature extraction, such as nonlinear kernel mapping [6] for spectral feature extraction and extended morphological profiles (EMPs) [7] for spatial feature extraction. For spectral–spatial feature learning, in [8], discriminant analysis and hypergraph learning were combined to represent the complex intrinsic spatial–spectral of HSI. Besides, Luo et al. [9] proposed a multistructure unified discriminative embedding method for learning the low-dimensional features of HSI, which obtained better classification results compared with other methods.

However, these traditional methods belong to "shallow" models and the ability of extracting high-level image features is restricted. Therefore, these methods usually fail to achieve higher accuracies [10]. In order to solve these problems, in the recent decade, deep learning models, which are data-driven and powerful for feature extraction [11], have dominated the classification of HSI [12], [13], [14]. Deep learning-based methods hierarchically extract the discriminant and invariant features of HSI and then obtain good results of HSI classification [15], [16].

In the early stage of deep learning-based HSI classification, traditional deep models, such as stacked autoencoder and deep belief network, were explored. Stacked autoencoders usually use several autoencoders to formulate a deep model to extract the nonlinear features of input. In [17], three stacked autoencoders have been designed to build spectral, spatial, and spectral–spatial HSI classifiers. Another deep model, i.e., deep belief network, was also introduced for HSI classification [18], [19]. However, stacked autoencoder and deep belief network used many fully connected layers, which usually contain a large number of learnable weights to be trained. In real applications, the number of HSI training samples is usually limited, and a lot of trainable weights in the deep model usually lead to a serious problem called overfitting.

In contrast to stacked autoencoder and deep belief network, convolutional neural network (CNN) uses local connection and

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shared weight. Local connection means that each neuron in a given convolutional filter is only connected to a part of the neurons of the previous layer and weight sharing ensures that the weights of a filter are fixed when processing the whole input [20]. Thus, the number of weights in CNN is significantly reduced [21]. Furthermore, CNN has powerful capability to extract the local features of the inputs. Due to these advantages, CNN-based methods have become the de facto standard for HSI classification [22], [23], [24], [25]. Many CNN-based methods have been proposed and they are roughly divided into three main categories: spectral feature extraction-based CNNs, spatial feature extraction-based CNNs, and spectral–spatial feature extraction-based CNNs.

For spectral feature extraction-based CNNs, the networks receive the pixel vector as the input and obtain the features of the hundreds of spectral bands, which are useful for the following classification [26]. Many works concentrated on exacting spectral features using 1-D convolutional kernels for HSI classification. For example, in [27], it was the first time that 1-D CNN was explored for HSI classification in spectral domain, and it obtained better classification results when compared with traditional methods.

Except for extracting spectral features with 1-D convolutional kernels, learning spatial features is also important for HSI classification [28]. For spatial feature extraction-based CNNs, 2-D convolutional kernels were used for HSI classification. For example, Jia and Lu [29] proposed an end-to-end classification method that designed an additional spatial constraint loss for the CNN's output. Also, some spatial feature extraction methods were combined with CNNs for better HSI classification performance, such as attribute profiles [30] and Haar wavelets [31].

Since the HSI is a 3-D cube with the spectral and spatial continuity, spectral–spatial feature extraction-based CNNs have gained good performance for HSI classification [32]. For instance, Yue et al. [33] proposed an adaptive spatial pyramid constraint method, which used 2-D convolutional kernels to extract global spatial neighborhood information and spectral information. Later, 3-D CNN has attracted lot of attention, which extracts spectral and spatial information at the same time. For example, in [34], 3-D convolutional kernels and residual blocks were explored for HSI classification and led to better performance. Besides patch-based networks, Zhu et al. [35] proposed a global learning framework, which combined convolutional long short-term memory and attention mechanism, to extract spectral–spatial features in the case of insufficient and imbalanced HSI training samples.

In general, CNN-based methods have achieved good performance for HSI classification. However, there are some difficulties in CNN-based HSI classifiers. CNNs are not good at sequential data processing. However, the spectra of HSI are sequential data with hundreds of bands [36], [37]. Moreover, CNNs use the local connection to extract features. Therefore, it is difficult to extract global features of the input. Recently, Transformer, which is a kind of relatively new deep learning model based on attention mechanism, has proved itself to be good at modeling long-range dependencies in the sequential data [10], [38]. Therefore, Transformer is investigated for HSI classification in this study. According to the architecture used for feature learning, most Transformer-based HSI classification methods can be summarized into the following two categories.

1) Methods that first use CNN and then use Transformer for spectral–spatial information processing. For example, Sun et al. [39] utilized a CNN to extract spectral–spatial features before tokenization for the following Transformer, which helped to generate more accurate token representations. The designed method in [40] used not only a spectral-adaptive 3-D convolution projection module first but also convolutional Transformers to deal with spectral–spatial features, which obtained comparative classification results.
2) Methods that solely use Transformer for spectral–spatial information processing. For example, in [41], the bidirectional encoder representations from transformers (BERT) model was explored for HSI classification, which flattened the pixels in a given region as an input sequence. In addition, Hong et al. [42] proposed a novel backbone network called SpectralFormer, which used multiple neighboring bands for one pixel or small block as the Transformer's sequential input.

However, it is acknowledged that Transformer is lack of local inductive bias. In machine learning, the term inductive bias refers to a set of assumptions that put some restrictions on the space of hypotheses in order to perform induction or prediction. Local inductive bias implies that closely placed pixels are related to each other [43]. Specifically, the neighboring pixels for an HSI image tend to be correlated and they usually belong to the same class, while distant pixels are usually less correlated. Different from CNN, the lack of inductive bias in modeling locality makes Transformer need a large amount of training data [44]. Especially for HSI classification task, the number of training samples is usually limited in practice because of the heavy annotation burden [45]. If the number of training samples is not sufficient, Transformer faces a serious generalization problem, which means that the model obtains good classification accuracy in the training stage but bad performance in the test stage.

To address this issue, we propose a spectral–spatial masked Transformer (SS-MTr) for HSI classification, which mainly pretrains a vanilla spectral–spatial Transformer model by masking some specific areas of the original input HSI images and predicting the missing pixels. Just like widely used cloze test in language learning and instruction, the model needs to focus on local spatial relationship among different Transformer patches when trying to restore the original HSI image from the masked image, which brings the inductive bias in modeling locality to the Transformer model and thus makes the model to be more data-efficient than the commonly used Transformers for HSI classification.

SS-MTr focuses more on learning the local semantic information, while the relationships among different HSI training samples are ignored. Therefore, the representations learned by pretrained SS-MTr are less discriminative, which is undesired for the downstream classification task. To solve this problem,

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contrastive learning (CL) and supervision label are combined with SS-MTr for embracing instance and interclass discriminability, respectively. Thus, three discriminative SS-MTrs (D-SS-MTrs) are then further proposed in this article, including contrastive SS-MTr (C-SS-MTr), supervised SS-MTr (S-SS-MTr), and supervised C-SS-MTr (SC-SS-MTr). Each of the D-SS-MTrs is extended from the base SS-MTr by adding extra branches for specific tasks in parallel with the existing reconstruction task. Generally, by integrating discriminative feature learning, masked samples reconstruction, and spectral–spatial vision Transformer, the proposed methods can obtain competitive classification performance.

The main contributions of this study are listed as follows.

1) A masked spectral–spatial Transformer-based learning framework, which injects the local inductive bias into Transformer via reconstruction from masked input HSI samples, is combined with multiscale data augmentation (MA) and spectral information aggregation (SIA) for HSI classification.
2) To increase instance discriminability, SS-Tr is combined with CL to form a new HSI classification method titled C-SS-MTr. C-SS-MTr uses the masked and the unmasked HSI inputs to generate plausible contrastive views. While maintaining modeling locality, the C-SS-MTr can learn instance discrimination information by distinguishing each HSI training sample from the others in the CL task.
3) To leverage class-level discriminability in the HSI samples, S-SS-MTr is proposed by adding a supervised classification branch to the SS-MTr. Benefiting from implicit supervision from the reconstruction task and explicit supervision from the classification task, the proposed S-SS-MTr not only models the local semantic information of the HSI samples but also captures the interclass dissimilarity and intraclass similarity.
4) Based on the C-SS-MTr and S-SS-MTr, SC-SS-MTr is proposed, which incorporates the reconstruction task, the CL task, and the supervised classification task in a unified framework. By learning local semantic-, instance-, and class-level features simultaneously, SC-SS-MTr improves the quality of learned representation in pretraining and thus benefits the final classification results.

The rest of this article is organized as follows. Section II presents the proposed SS-MTr for HSI classification. Section III describes the three proposed D-SS-MTrs, including C-SS-MTr, S-SS-MTr, and SC-SS-MTr. Section IV presents the HSI classification results and analysis of comprehensive experiments. In Section V, the article's conclusion is briefly summarized.

II. Proposed SS-MTr for HSI Classification

Motivated by the recent works on pretraining frameworks in computer vision and Transformer-based models for HSI classification, this study explores SS-MTr for HSI classification. The proposed SS-MTr pretrains a vanilla Transformer model by masking some specific areas of the original input HSI image and predicting the missing pixels. The overall training flow of SS-MTr is shown in Fig. 1. Generally, the training process for HSI classification entails two stages: pretraining stage and fine-tuning stage.

In the pretraining stage, the HSI sample is first augmented by MA, which is used to reduce the risk of overfitting and help Transformers to learn the spatial structural details of an input. Then, the augmented HSI input is masked and sent to a spectral–spatial Transformer model, which consists of an SIA module and a Transformer-based encoder, for spectral–spatial feature extraction. Next, a Transformer-based decoder is used to reconstruct the original HSI input from the extracted features. Finally, a reconstruction loss between the reconstructed and original inputs is computed for model updating.

In the fine-tuning stage, a fully connected layer is added to the pretrained Transformer encoder, which maps the class token to logits for classification. This network will then be fine-tuned by using cross-entropy loss to accomplish the final HSI classification task.

A. Spatial–Spectral Transformer for HSI Feature Extraction

Transformer has proved itself to be good at modeling long-range dependencies in data. Therefore, in this section, we mainly focus on a vanilla spectral–spatial Transformer for HSI classification (SS-Tr), which is the backbone of the proposed SS-MTr-based methods in the following.

A standard vision Transformer processes an input image as if it was a sequence of input tokens, which are the basic processing units for a Transformer-based encoder [46]. To achieve this purpose, the input image is spatially divided into several nonoverlapping patches. Then, these patches are linearly projected into a sequence of patch tokens, which are going to be processed by the Transformer-based encoder for feature extraction.

Specifically, a Transformer encoder consists of several blocks, each of which involves alternating layers of multihead self-attention (MSA), multilayer perceptron (MLP), and layer normalization (LN) [47], as shown in Fig. 1.

The output tokens $\mathbf{Z}^{(l)}$ for the lth block are obtained as follows:

$$\tilde{\mathbf{Z}}^{(l)} = \mathrm{MSA}(\mathrm{LN}(\mathbf{Z}^{(l-1)})) + \mathbf{Z}^{(l-1)} \tag{1}$$

$$\mathbf{Z}^{(l)} = \mathrm{MLP}(\mathrm{LN}(\tilde{\mathbf{Z}}^{(l)})) + \tilde{\mathbf{Z}}^{(l)} \tag{2}$$

where $\mathrm{LN}(\cdot)$ denotes the LN operation, $\mathrm{MSA}(\cdot)$ denotes the mapping function of MSA, and $\mathrm{MLP}(\cdot)$ represents the MLP layer. Also, $\mathbf{Z}^{(l-1)}$ means the output tokens for the previous block.

However, it is not suitable to directly apply the standard vision transformer for HSI feature extraction. HSI is a 3-D cube with the spectral and spatial continuity, but the standard vision Transformer, which is based on spatial patch partition, mainly focuses on spatial self-attention. In order to fully exploit the spectral–spatial information contained in HSI, we use an SIA module to condense spectral features before performing spatial partition for the Transformer encoder, leading to the spectral–spatial Transformer for HSI feature extraction.

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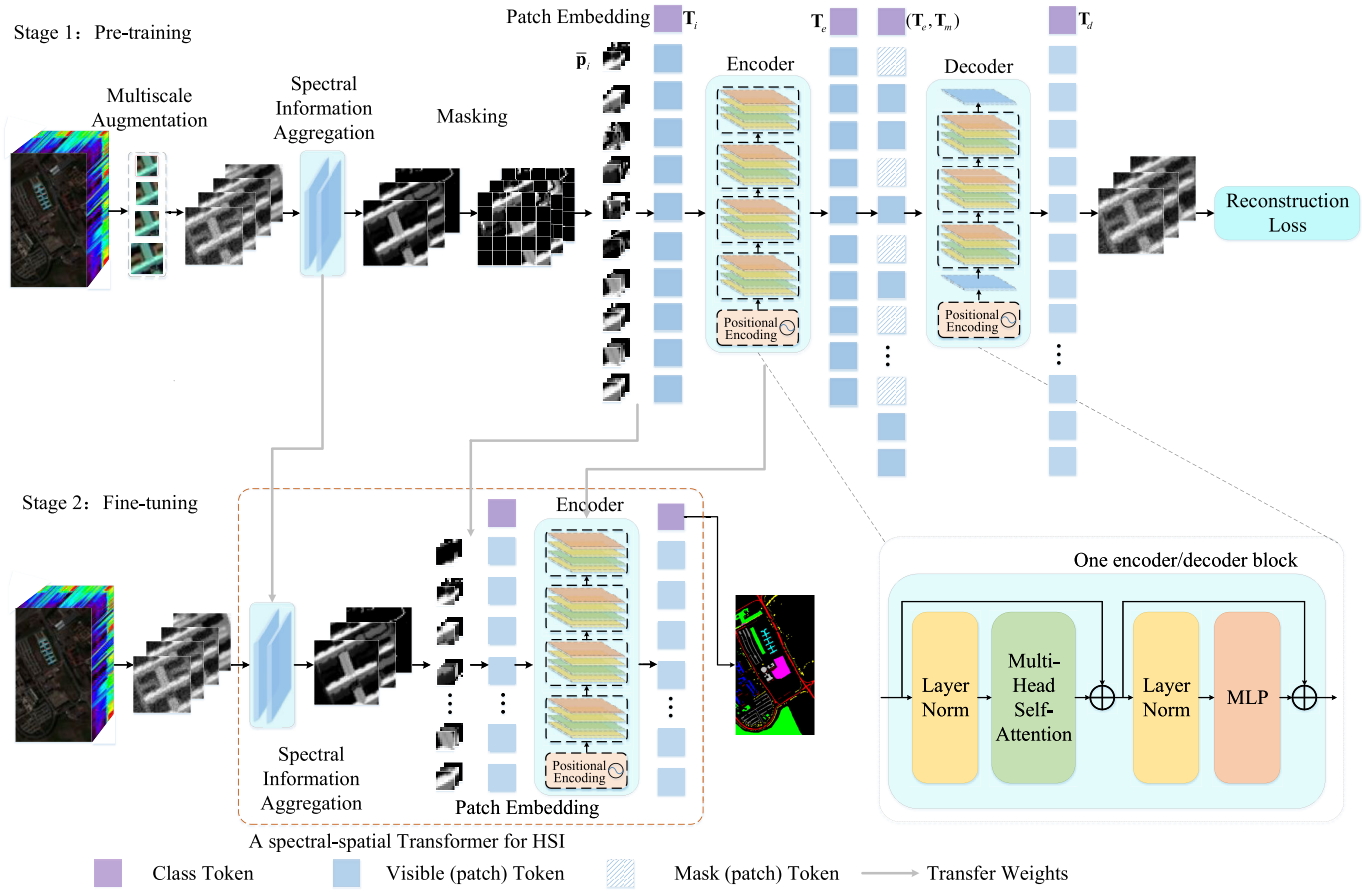


Fig. 1. Proposed SS-MTr for HSI classification overview—training flow. For illustrative purposes, a single image flow instead of a batch is shown here. Stage 1: pretraining for the reconstruction task. Stage 2: fine-tuning for the final classification task. In the pretraining stage, the patch embedding matrix, the SIA module, the encoder, and the decoder are trained via the reconstruction task. After pretraining, the SIA module, the patch embedding matrix, and the encoder are left unchanged for the downstream classification task in the fine-tuning stage.

Given an HSI training sample $\mathbf{x} \in \mathbb{R}^{H\times W\times B}$, where H and W are the height and width of the input, respectively, and B is the number of spectral bands, the SIA linearly combines the images of each band to obtain the output $\bar{x} \in \mathbb{R}^{H\times W\times C}$

$$\bar{x}_{i,j,c} = \sum_{m=1}^{B} \alpha_{m,c} x_{i,j,m} \tag{3}$$

where $\boldsymbol{\alpha}$ are learnable parameters for optimization and C is the number of transformed bands. Equation (3) shows that each band of the input HSI sample is multiplied by learnable weights, and then, they are added together to form a new 3-D HSI input. The procedure of SIA can be achieved by linear transformation and is denoted as

$$\bar{x} = \mathrm{SIA}(\mathbf{x}) \tag{4}$$

where $\mathrm{SIA}(\cdot)$ represents the SIA mapping function.

Then, the spectrally aggregated sample $\bar{x}$ is sent to a standard vision Transformer, which has been introduced before, for long-range spectral–spatial feature extraction.

B. SS-MTr for HSI Classification

The lack of inductive bias in modeling locality makes Transformer need a large amount of training data, which is not always satisfied because of the heavy annotation burden in HSI training data preparing. To address this issue, we propose the SS-MTr for HSI classification in this section.

As shown in Fig. 1, in the pretraining stage, SS-MTr maps a masked HSI training sample to latent representations and uses a decoder to reconstruct the original signals. The reconstruction task based on the given partial information enables the encoder to learn the local spatial relationship among different image patches.

Given the spectrally aggregated HSI sample $\bar{x}$ introduced in Section II-A, we first divide it into $N = HW/P^2$ nonoverlapping patches $\bar{\mathbf{p}}$, where $\bar{\mathbf{p}} \in \mathbb{R}^{N\times P^2 C}$ and P is the spatially partitioned patch size.

Then, a random masking method with respect to image patches is adopted, which randomly selects a subset of all patch tokens following a uniform distribution without replacement. Specifically, let I denote the set of patch tokens' indexes

$$I = \{a | 1 \le a \le N, a \in Z\} \tag{5}$$

where Z denotes the set of integers. Then, the indexes for masked patches I_m are generated by

$$I_m = \mathrm{RandomSelect}(I, Np) \tag{6}$$

where $\mathrm{RandomSelect}(I, Np)$ means selecting Np elements from the given set I without replacement and p is the masking

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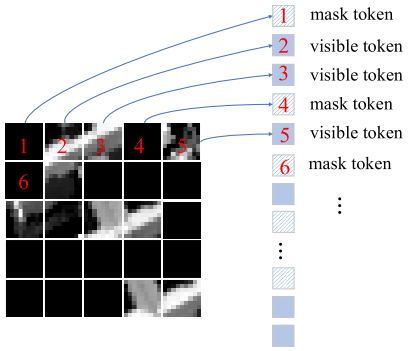


Fig. 2. Illustration of visible token and mask token.

ratio. Also, a masked view of $\bar{x}$ can be obtained using the selected indexes

$$\bar{\mathbf{p}}_i = \bar{\mathbf{p}}[I - I_m] \in \mathbb{R}^{N(1-p)\times P^2C}. \tag{7}$$

After that, the left patches $\bar{\mathbf{p}}_i$ are linearly projected into a sequence of patch tokens by means of a learnable matrix $\mathbf{E} \in \mathbb{R}^{P^2C\times D}$, where D represents the tokens' dimension for encoder

$$\mathbf{T}_i = [\mathbf{t}_{\text{cls}}; \bar{\mathbf{p}}_i\mathbf{E}] \tag{8}$$

where $\mathbf{t}_{\text{cls}}$ is another learnable vector named the class token appended to the patch tokens and $\mathbf{T}_i \in \mathbb{R}^{(N-Np+1)\times D}$ denotes the obtained tokens.

As shown in Fig. 1, the positional encoding matrix $\mathbf{H} \in \mathbb{R}^{N\times D}$ is then employed to provide location information for tokens

$$\begin{cases} H_{i,2j} = \sin\left(\dfrac{i}{1000^{\frac{2j}{D}}}\right) \\ H_{i,2j-1} = \cos\left(\dfrac{i}{1000^{\frac{2j}{D}}}\right) \end{cases} \quad j = 1, 2, \ldots, D/2. \tag{9}$$

Tokens $\mathbf{T}_i$ are then updated by adding with masked encoding matrix $\mathbf{H}[I - I_m]$. Also, they are fed into a Transformer-based encoder (L_e Transformer blocks), which has been introduced before, for spectral–spatial features extraction

$$\mathbf{T}_e = TE(\mathbf{T}_i) \tag{10}$$

where $\mathbf{T}_e \in \mathbb{R}^{(N-NP+1)\times D}$ denotes the obtained tokens and TE$(\cdot)$ represents the mapping function of the encoder.

Next, a Transformer-based decoder, which is composed of L_d Transformer blocks and two linear layers, is used to reconstruct the original HSI sample. The input tokens for the decoder come from: 1) encoded tokens $\mathbf{T}_e$, which contain the partial context information of an HSI sample, and 2) mask tokens $\mathbf{T}_m$, which serve to complete the full input sample and indicate the presence of missing patches. Each mask token is a learnable vector that is initialized with $\mathbf{0}$. The masked tokens are added with the encoded tokens $\mathbf{T}_e$ according to their original positions, just as shown in Fig. 2.

The first layer of the decoder is a linear layer, which transforms the input tokens' dimension from D to D'. Then, the transformed tokens are processed by the Transformer blocks. The position encoding operation is also performed to provide location information for the reconstruction task. To reconstruct the original HSI sample, the obtained tokens are then linearly projected to the pixel space by another linear layer. The general process of the decoder can be represented by

$$\mathbf{T}_d = \text{TD}(\mathbf{T}_e, \mathbf{T}_m) \tag{11}$$

where $\mathbf{T}_d \in \mathbb{R}^{(N+1)\times P^2B}$ denotes the obtained tokens used for reconstruction and TD$(\cdot)$ represents the mapping function of the decoder. Note that the obtained $\mathbf{T}_d$ can be reshaped to an image, which shares the same shape with the original HSI sample $\mathbf{x}$, after removing the class token.

Finally, the mean squared error between the reconstructed and original HSI samples is computed only at the masked area. For simplicity, image patches rather than image cubes are used for computing loss. Specifically, the original HSI sample $\mathbf{x}$ is also spatially divided into N nonoverlapping patches $\mathbf{p}$, and then, the reconstruction loss $\mathcal{L}_{\text{rec}}$ is calculated by

$$\tilde{\mathbf{p}} = \mathbf{T}_d[1 : \text{end}] \tag{12}$$

$$\mathcal{L}_{\text{rec}} = \left\|\mathbf{p}[I_m] - \tilde{\mathbf{p}}[I_m]\right\|_2^2. \tag{13}$$

For better understanding the SS-MTr, Table I shows the shapes of feature maps after processed by different layers in different stages. The encoder has four blocks, whose output feature maps' shapes are all $(N - Np + 1) \times D$, and the decoder has two linear layers, whose feature maps' shape is $(N + 1) \times D'$ and $(N + 1) \times P^2B$. The decoder has also three Transformer blocks, whose output feature maps' shapes are all $(N + 1) \times D'$. In the finetuning stage, the encoder is reserved without masking, leading to the fact that the output feature maps' shapes for each block are all $(N + 1) \times D$.

TABLE I
SHAPES OF FEATURE MAPS FOR EACH LAYER

Layer	Pre-training stage		Fine-tuning stage
	encoder	decoder	encoder
Linear Layer 1	-	$(N + 1) \times D'$	-
Block 1	$(N - Np + 1) \times D$	$(N + 1) \times D'$	$(N + 1) \times D$
Block 2	$(N - Np + 1) \times D$	$(N + 1) \times D'$	$(N + 1) \times D$
Block 3	$(N - Np + 1) \times D$	$(N + 1) \times D'$	$(N + 1) \times D$
Block 4	$(N - Np + 1) \times D$	-	$(N + 1) \times D$
Linear Layer 2	-	$(N + 1) \times P^2B$	-

SS-MTr uses the neighborhood region surrounding a given pixel to generate the input HSI sample, which is intuitive and prevalent. However, due to the spatial partition operation, a specific pixel in the input sample is assigned to a patch, which will not change during the whole process and may cause spatial information loss. Besides, in the HSI image, different ground objects usually have various sizes and shapes [48]. Especially for the limited training samples, the model memorizes lots of specific information of land covers, most of which is unilateral if using single fixed scale, which finally leads to overfitting. Therefore, MA, which can not only help the Transformer learn the detailed spatial information of HSI image but also increase the diversity of training samples to reduce the risk of overfitting, is also explored in the proposed

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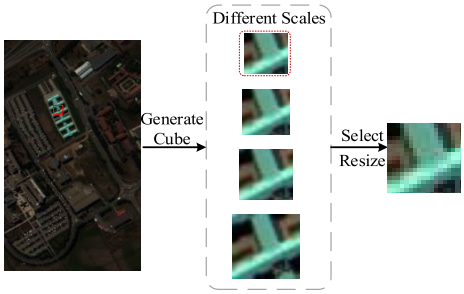


Fig. 3. MA.

Algorithm 1 Pseudocode of Pretraining for SS-MTr

Input: input training sample's location loc, HSI scene $\mathbf{X}$, minimum size S, mask ratio p.
Initialization: embedding matrix $\mathbf{E}$, networks SIA, TE, and TD
Output: embedding matrix $\mathbf{E}$, pre-trained networks SIA and TE
$\mathbf{x} = MA(\mathbf{X}, S, loc)$
$\bar{x} = SIA(\mathbf{x})$
get image patches $\bar{\mathbf{p}}$
get visible patches $\bar{\mathbf{p}}_i$ using Eq. (5), Eq. (6) and Eq. (7)
get input tokens $\mathbf{T}_i$ for the encoder using Eq. (8)
$\mathbf{T}_e = TE(\mathbf{T}_i)$
$\mathbf{T}_d = TD(\mathbf{T}_e, \mathbf{T}_m)$
Compute reconstruction loss using Eq. (12) and Eq. (13) back-propagation and update $\mathbf{E}$, SIA, TE, and TD
return $\mathbf{E}$, SIA and TE

SS-MTr. Specifically, the MA randomly selects a size (e.g., 21×21) from several candidate sizes (e.g., 27×27, 25×25, $\ldots$, $S \times S$, where S denotes the minimum size) for any given sample. Then, an HSI sample with the selected spatial size, which is a neighborhood region centered by the given training pixel, is generated from the whole HSI scene $\mathbf{X}$. Finally, the sample is resized to the desired spatial size (i.e., $H \times W$) using bilinear interpolation. The MA procedure can be summarized as

$$\mathbf{x} = \mathrm{MA}(\mathbf{X}, S, \mathrm{loc}) \tag{14}$$

where $\mathrm{MA}(\cdot)$ represents the mapping function of MA, $\mathbf{x}$ denotes the obtained HSI sample, and loc denotes the location of the given pixel. The procedure of MA is shown in Fig. 3.

As shown in Fig. 1, the SIA module, the patch embedding matrix, and the encoder are left unchanged for the downstream classification task in the fine-tuning stage. Specifically, a fully connected layer is added following the encoder to form a new network for HSI classification. Also, the original HSI samples, i.e., without MA and masking, is used for fine-tuning the classification network. Algorithm 1 demonstrates the pretraining procedure for SS-MTr.

III. Proposed D-SS-MTrs for HSI Classification

The SS-MTr focuses more on learning the local semantic information of the HSI samples via the reconstruction task, while the relations among different HSI training samples are neglected. Therefore, the representations learned by SS-MTr are less discriminative, which can be shown in the experiments. However, the downstream classification task mainly relies on discriminative information across the training HSI samples. Based on the above analysis, one can ask: can we incorporate discriminative feature learning into SS-MTr's pretraining?

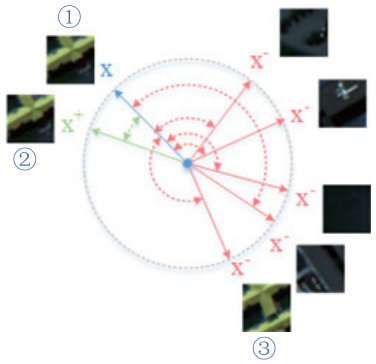


Fig. 4. Illustrations of instance discriminability learning in the representation space. The green dashed lines mean pulling toward each other, whereas the red dashed lines mean pushing away from each other.

To answer the above question, this study proposes D-SS-MTrs in two ways in this section, including CL, which focuses on instance discriminability, and supervision of labels, which focuses on interclass discriminability. Instance discriminability means the ability to discriminate one instance (i.e., an HSI sample) from the other instances, regardless of whether they belong to the same class [49]. By comparison, the interclass discriminability emphasizes the difference of features between distinct classes. Taking image classification as an example, an image is distinctive in its own right, and each can differ significantly from other images in the same semantic category. The class-level discriminability emphasizes the difference of features belonging to distinct classes, while the individual difference is emphasized for the instance-level discriminability, that is, even the features of the same class can preserve individual differences to a certain extent while maintaining clustering properties. As shown in Fig. 4, images 1 and 2 are different augmented views for the same sample, while image 3 is another sample from the same class with images 1 and 2. Only the features from the same sample are pulled toward each other in the representation space. The pretrained features with better instance or interclass discriminability are thought beneficial for the downstream tasks.

Based on these two ways, three SS-MTr-based HSI classification methods, which can effectively learn discriminative features, are developed.

A. C-SS-MTr for HSI Classification

CL, which learns representations by maximizing agreement between augmented views of the same sample in the latent space and maximizing difference for two distinct samples, is a well-proven and prevailing method for neural network pretraining. It is universally acknowledged that the pretrained model is endowed with strong instance discriminability by a CL strategy. Therefore, combining CL and SS-MTr, i.e.,

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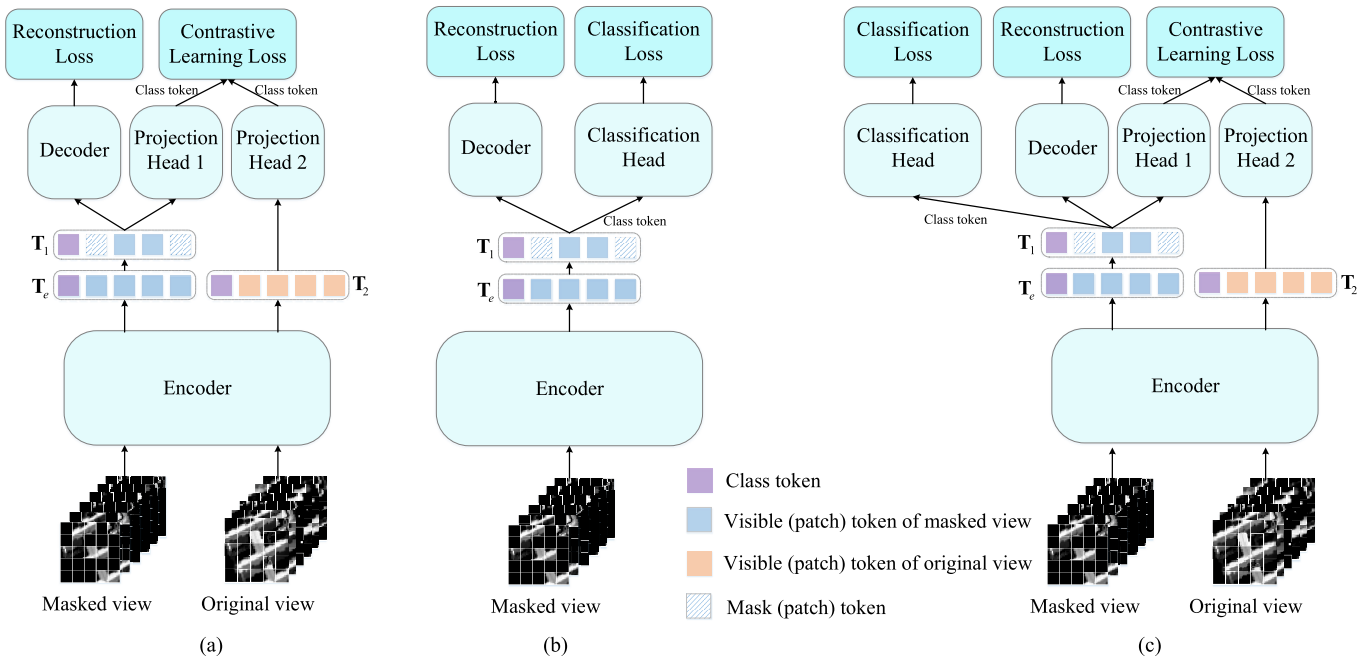


Fig. 5. Proposed D-SS-MTrs for HSI classification overview—pretraining flow. For illustrative purposes, a single image flow instead of a batch is shown here. Each of the D-SS-MTrs is extended from the base SS-MTr by adding extra branches for specific tasks in parallel with the existing reconstruction task. (a) C-SS-MTr. (b) S-SS-MTr. (c) SC-SS-MTr.

C-SS-MTr, is proposed in this section for strengthening the representations capability.

An overview of the proposed C-SS-MTr is shown in Fig. 5(a). In general, C-SS-MTr is extended from SS-MTr by adding a branch for CL in parallel with the existing reconstruction branch.

Two correlated views of any given HSI training sample are usually needed in CL. In C-SS-MTr, it is convenient and intuitive to take the masked HSI sample and the unmasked HSI sample as two views for the given training sample.

For the masked view $\bar{\mathbf{p}}_i$, the corresponding tokens, which are composed of encoded tokens and mask tokens [i.e., $(\mathbf{T}_e, \mathbf{T}_m)$], are denoted as $\mathbf{T}_1$. For the unmasked view $\bar{\mathbf{p}}$, the tokens are denoted as $\mathbf{T}_2$, which can be generated by

$$\mathbf{T}_2 = TE([\boldsymbol{t}'_{\text{cls}}; \bar{\mathbf{p}}\mathbf{E}]) \tag{15}$$

where $\boldsymbol{t}'_{\text{cls}}$ is the class token for the unmasked view.

For the contrastive leaning task, each of the two sequences of tokens $\mathbf{T}_1$ and $\mathbf{T}_2$ is fed into a corresponding projection head, which is made up of one Transformer block. Also, the outputs are denoted by $\mathbf{Z}_1$ and $\mathbf{Z}_2$. The classification tokens of $\mathbf{Z}_1$ and $\mathbf{Z}_2$, denoted by $\mathbf{z}_{\text{cls}}$ and $\mathbf{z}'_{\text{cls}}$, are then used to compute the CL loss for model updating.

The CL loss adopts the form of InfoNCE loss function, which is computed across different samples in any given mini-batch [50]. Some notations must be clarified first. Suppose that there are M training samples in in one mini-batch, and $\{\mathbf{h}_i\}_{i=1}^{2M}$ is used to denote the classification tokens for all the M samples

$$\begin{cases} \mathbf{h}_{2k} = \mathbf{z}_{\text{cls}}^{[k]} \\ \mathbf{h}_{2k-1} = \mathbf{z}_{\text{cls}}^{[k]'} \end{cases} \quad k = 1, 2, \ldots, M \tag{16}$$

where $\mathbf{z}_{\text{cls}}^{[k]}$ and $\mathbf{z}_{\text{cls}}^{[k]'}$ are the classification tokens of two views for the kth samples. Let $s(\mathbf{a}, \mathbf{b}) = \mathbf{a}^T\mathbf{b}/(\|\mathbf{a}\|\|\mathbf{b}\|)$ represent the inner product between ℓ_2 normalized $\mathbf{a}$ and $\mathbf{b}$. Then, the contrastive loss function is defined as

$$\mathcal{L}_{\text{cl}} = \frac{1}{2M}\sum_{k=1}^{M}(\ell(2k-1, 2k) + \ell(2k, 2k-1)) \tag{17}$$

$$\ell(i, j) = -\log\frac{e^{s(\mathbf{h}_i, \mathbf{h}_j)}}{\sum_{k=1}^{2M}\mathbb{1}_{[k\neq i]}e^{s(\mathbf{h}_i, \mathbf{h}_k)}} \tag{18}$$

where $\mathbb{1}_{[\cdot]} \in \{0, 1\}$ is an indicator function returning 1 if and only if the condition in $[\cdot]$ is satisfied.

Meanwhile, the tokens $\mathbf{T}_1$ are further processed by the decoder for the reconstruction task, and the reconstruction loss $\mathcal{L}_{\text{rec}}$ is calculated.

The total loss function for C-SS-MTr is defined as a weighted sum of the above two loss functions

$$\mathcal{L}_{\text{rc}} = \mathcal{L}_{\text{rec}} + \gamma\mathcal{L}_{\text{cl}} \tag{19}$$

where γ is a balancing coefficient to weight the importance of CL loss.

B. S-SS-MTr for HSI Classification

Using supervision labels to SS-MTr is another direct but effective approach to obtain strong discriminability for HSI classification. Different from instance discriminability by CL strategy, the supervision of labels endows the SS-MTr with interclass dissimilarity and intraclass similarity. It is more suitable for the downstream HSI classification task.

An overview of the proposed S-SS-MTr is shown in Fig. 5(b). Like the proposed C-SS-MTr, S-SS-MTr is extended from SS-MTr by adding a branch for supervised classification in parallel with the existing reconstruction branch.

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The classification token of the encoder's output is used for an extra classification task. Specifically, a classification head, which is made up of several fully connected layers together with ReLU activation function and batch normalization layer, is used to map the classification token to prediction $\hat{\mathbf{y}} \in \mathbb{R}^K$, where K is the number of classes. Cross-entropy loss is used here to compute the difference between predicted and true labels for any given training sample

$$\mathcal{L}_{\text{cls}} = -\sum_{k=1}^{K} y_k \log \hat{y}_k \tag{20}$$

where y_k is the kth element of the one-hot label vector.

S-SS-MTr learns representations via both the reconstruction task and classification task. Also, the total loss function is defined as a weighted sum of these two loss functions

$$\mathcal{L}_{rs} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{cls}} \tag{21}$$

where λ is a balancing coefficient to weight the importance of classification loss.

C. SC-SS-MTr for HSI Classification

In this section, both CL and supervision labels are incorporated into SS-MTr to develop a new D-SS-MTr for HSI classification. The proposed SC-SS-MTr has the ability of not only modeling the local semantic information of the HSI samples but also capturing the relation of different HSI samples and even different classes. Meanwhile, the CL can serve as an additional regularization for classification task, which helps the training process and alleviates the risk of overfitting.

An overview of the proposed SC-SS-MTr is shown in Fig. 5(c). There are three learning tasks: 1) the masked view and the Transformer decoder are served for the reconstruction task; 2) the masked view and the classification head are served for the classification task; and 3) the masked view, the original view, and the corresponding projection heads are served for the CL task. All the components related to the SC-SS-MTr have been introduced in Sections III-A and III-B.

The total loss function for SC-SS-MTr is defined as

$$\mathcal{L}_{\text{rsc}} = L_{\text{rec}} + \lambda \mathcal{L}_{\text{cls}} + \gamma \mathcal{L}_{\text{cl}} \tag{22}$$

where $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{cl}}$ are the classification loss and CL loss, respectively, and λ and γ are the corresponding balancing coefficients introduced in (19) and (21).

It is worth noting that both SS-MTr and C-SS-MTr are unsupervised HSI pretraining methods, while S-SS-MTr and SC-SS-MTr are both supervised HSI pretraining methods.

IV. Experimental Results and Discussion

A. Dataset Description

In the experiments, the performance of the proposed methods is evaluated on three widely used hyperspectral datasets, including Indian Pines, Pavia University, and Houston.

1) Indian Pines: Indian Pines dataset mainly covers the scene of agricultural fields in Northwestern Indiana, USA. It was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in June 1992. The dataset contains 145 × 145 pixels and 220 spectral bands with wavelengths ranging from 400 to 2500 nm. The spatial resolution is 20 × 20 m. In the experiments, 200 bands are used for evaluation after removing 20 low signal-to-noise ratio (SNR) bands. This dataset has 10 249 labeled samples and 16 different land-cover types. Fig. 6 shows the false-color map and ground-truth map of the Indian Pines dataset. Table II lists the numbers of the training and test samples per class.



Fig. 6. Indian Pines dataset: (a) false-color map and (b) ground truth.

TABLE II
Land Cover Classes and Numbers of Samples for the Indian Pines Dataset

No.	Class Name	Training samples	Test samples	Total samples
1	Alfalfa	20	26	46
2	Corn-notill	20	1408	1428
3	Corn-mintill	20	810	830
4	Corn	20	217	237
5	Grass-pasture	20	463	483
6	Grass-trees	20	710	730
7	Grass-pasture-mowed	20	8	28
8	Hay-windrowed	20	458	478
9	Oats	15	5	20
10	Soybean-notill	20	952	972
11	Soybean-mintill	20	2435	2455
12	Soybean-clean	20	573	593
13	Wheat	20	185	205
14	Woods	20	1245	1265
15	Buildings-Grass-Trees	20	366	386
16	Stone-Steel-Towers	20	73	93
	Total	315	9934	10249

2) Pavia University: This dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the University of Pavia, Italy. This dataset contains 610 × 340 pixels and 103 spectral bands. The spatial resolution is 1.3 × 1.3 m. The ground truth comprises 42 776 labeled pixels belonging to nine classes. The false-color map and the ground-truth map of the Pavia University dataset are shown in Fig. 7. The numbers of the training and test samples per class are listed in Table III.

3) Houston: This dataset describes an urban area about the University of Houston campus in USA, acquired by the National Center for Airborne Laser Mapping (NCALM). It had been used for the 2013 IEEE GRSS Data Fusion Contest [51]; 349 × 1905 pixels with a spatial resolution of 2.5 × 2.5 m are contained in this dataset. The scene consists of 144 bands with wavelengths ranging from 380 to 1050 nm. The 15 029

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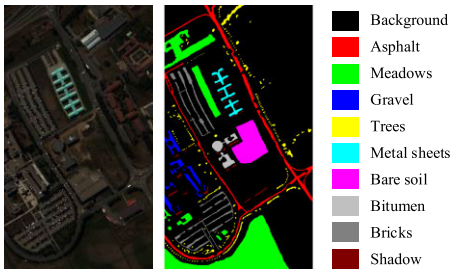


Fig. 7. Pavia University dataset: (a) false-color map and (b) ground truth.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Class Name	Training samples	Test samples	Total samples
1	Asphalt	20	6611	6631
2	Meadows	20	18629	18649
3	Gravel	20	2079	2099
4	Trees	20	3044	3064
5	Mental sheets	20	1325	1345
6	Bare soil	20	5009	5029
7	Bitumen	20	1310	1330
8	Bricks	20	3662	3682
9	Shadow	20	927	947
	Total	180	42596	42776

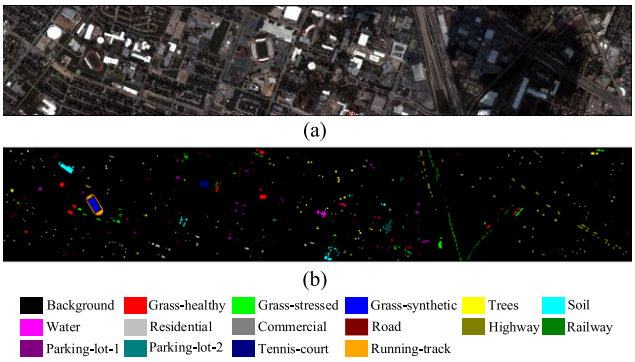


Fig. 8. Houston dataset: (a) false-color map and (b) ground truth.

labeled pixels belonging to 15 different land-cover types are collected in the dataset. Fig. 8 shows the false-color map and the corresponding ground-truth map. The numbers of the training and test samples per class are listed in Table IV.

B. Experimental Setup

The proposed SS-MTr-based methods are compared with several HSI classification methods to demonstrate the effectiveness. These comparison methods are designed as follows.

1) *EMP-SVM [7]:* This method uses EMP for spatial feature extraction and SVM for final classification. It serves as a traditional method for comparison with deep learning-based methods.
2) *SuTabNets [52]:* It uses structure profile, unsupervised pretraining, spatial attention, and tabular learning to extract spectral–spatial features.

TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE HOUSTON DATASET

No.	Class Name	Training samples	Test samples	Total samples
1	Grass-healthy	20	1231	1251
2	Grass-stressed	20	1234	1254
3	Grass-synthetic	20	677	697
4	Tree	20	1224	1244
5	Soil	20	1222	1242
6	Water	20	305	325
7	Residential	20	1248	1268
8	Commercial	20	1224	1244
9	Road	20	1232	1252
10	Highway	20	1207	1227
11	Railway	20	1215	1235
12	Parking-lot-1	20	1213	1233
13	Parking-lot-2	20	449	469
14	Tennis-court	20	4008	428
15	Running-track	20	640	660
	Total	300	14729	15029

3) *CNN:* It is a vanilla CNN with four convolutional layers, which is a basic spectral–spatial deep model for HSI classification. It is used here to be compared with SS-Tr to demonstrate the weakness of Transformer in the case of limited training samples.
4) *FDSSC [53]:* It is a fast dense spectral–spatial convolution network, which uses different convolutional kernel sizes to extract spectral and spatial features separately. In particular, FDSSC is an improved CNN, which can achieve competitive accuracies.
5) *DBDA [54]:* It is an improved CNN that uses double-branch dual-attention mechanism for more discriminative feature extraction. DBDA is based on attention mechanism and it is compared with Transformer-based methods, which is based on self-attention mechanism.
6) *MSSG [55]:* It uses a superpixel structured graph U-Net to learn multiscale features on multilevel graphs. As graph convolutional and global-based methods, it is compared with the proposed Transformer and patch-based methods.
7) *SSFTT [39]:* It is a spatial–spectral Transformer that utilized the advantages of hybrid CNN and Transformer. It is used to demonstrate the superiority over other Transformer-based methods.

Generally, the comparison methods cover traditional method, tabular learning method, CNN-based methods, Transformer-based methods, attention-based methods, graph convolution, and global method, which are complete and sufficient for comparison.

For EMP-SVM, the grid search strategy is utilized together with fivefold cross validation to find the proper C and $\gamma\left(C = 10^{-4}, 10^{-3}, \ldots, 10^{3}, \gamma = 10^{-4}, 10^{-3}, \ldots, 10^{3}\right)$. The first four principle components (PCs) are used when calculating EMP. For each PC, three openings and closings by reconstruction are conducted with a circular structuring

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TABLE V
ARCHITECTURE OF CNN

No.	Convolution	ReLU	Pooling	Padding	Stride	BN	Linear
1	1×1×32	YES	No	NO	1	YES	-
2	4×4×32	YES	2×2	NO	1	YES	-
3	3×3×64	YES	2×2	NO	1	YES	-
4	4×4×128	YES	2×2	NO	1	YES	-
5	-	YES	NO	NO	-	NO	128×256
6	-	NO	NO	NO	-	NO	256×K

element whose initial size is four and step size increment is two.

For SuTabNets, DBDA, FSDCC, MSSG, and SSFTT, the experimental settings are the same as their original papers. For CNN, the detailed architecture is shown in Table V.

For the Transformer-based encoder used in the proposed methods, four blocks are used ($L_e = 4$), each of which contains an eight-head self-attention layer, two LN layers, and an MLP whose hidden dimension is set to be as twice as the tokens' dimension for encoder (i.e., $D = 128$). In order to use spatial information, the input images with spatial size of 27 × 27 are fed to these models. Also, the patch size P is set to 3.

The Adam optimizer is used for pretraining SS-MTr-based methods, including SS-MTr, C-SS-MTr, S-SS-MTr, and SC-SS-MTr, with an initial learning rate of 0.001. Also, the cosine annealing with warm restart for learning rate adjusting is adopted in the experiments.

When training CNN, SS-Tr, and fine-tuning SS-MTr-based methods, the multistep learning rate scheduler is utilized. The initial learning rate is 0.001, and it is divided by ten after 80 and 160 epochs. During the training procedure, the total number of epochs is set to 180 for the three datasets. All the designed methods adopt mini-batch strategy with a batch size of 512.

Some hyperparameters (e.g., the minimum augmentation scale S and the masking ratio p) in the proposed SS-MTr-based methods have influence on the classification performance to a certain extent and they will be analyzed in the experiments.

In the experiments, the classification performance is mainly evaluated using overall accuracy (OA), average accuracy (AA), and Kappa coefficient (K). Experiments are repeated ten times with respect to different random initializations. Also, the training samples are randomly chosen from all labeled samples each time.

C. Classification Results

The proposed methods together with comparison methods are evaluated on the three datasets. Also, Tables VI–VIII show the classification results. It can be observed that the proposed S-SS-MTr and SC-SS-MTr are superior to the comparison methods with 20 training samples per class. Specifically, it can be seen that the following conditions hold.

1) The SS-Tr obtains worse classification performance than the CNN with respect to OA, AA, and K on the three datasets, which shows the weakness of Transformer with limited training samples. For example, in Table V, the CNN outperforms SS-Tr by 7.7 percentage points, 4.91 percentage points, and 0.0861 in terms of OA, AA, and K on the Indian Pines dataset, respectively.

2) The classification accuracies obtained by SS-Tr are higher than the ones obtained by SS-Tr, which shows the effectiveness of pretraining by reconstruction task. In Table V, compared to the SS-Tr, SS-MTr improves the classification accuracies by 1.79 percentage points, 1.42 percentage points, and 0.02 in terms of OA, AA, and K on the Indian Pines dataset, respectively. However, the SS-MTr can still not achieve competitive classification performance on the three datasets, when compared with methods such as FDSSC and SSFTT.

3) The incorporation of CL and supervision labels boosts the classification performance of SS-MTr. On one hand, the use of CL further improves the classification performance, which can be shown from the fact that the OA, AA, and K obtained by C-SS-MTr are higher than the ones obtained by SS-MTr. For example, in Table VI, the C-SS-MTr outperforms SS-MTr by 4.1 percentage points, 3.37 percentage points, and 0.0519 in terms of OA, AA, and K on the Pavia University dataset, respectively. On the other hand, the use of label supervision brings great gains to SS-MTr. The classification accuracies obtained by S-SS-MTr are higher than SS-MTr, C-SS-MTr, and other state-of-the-art methods. For example, in Table VII, the OA obtained by S-SS-MTr on the Houston dataset outperforms the ones of FDSSC, SSFTT, and C-SS-MTr by 0.36 percentage points, 1.45 percentage points, and 3.9 percentage points, respectively.

4) The SC-SS-MTr obtains the highest accuracies when compared with other methods on the three datasets, which demonstrates the effectiveness for the combination of reconstruction task, CL task, and the supervised classification task. For example, in Table VII, the OA obtained by SC-SS-MTr on the Houston dataset outperforms the ones of FDSSC, SSFTT, and S-SS-MTr by 0.91 percentage points, 2.00 percentage points, and 0.55 percentage points, respectively.

D. Ablation Experiments

1) Ablation Experiment With SIA: Table IX shows the classification results obtained by Transformer-based methods with or without SIA. It can be seen that the SIA is so important that the accuracies obtained by Transformer-based methods are significantly low if SIA is not used. We can infer that dealing with spectral features first is vital for spatial partition-based vision Transformers.

2) Ablation Experiment With the Reconstruction Target: The reconstruction loss is computed between the reconstructed sample and original HSI sample only at the masked area. Table X shows the ablation experimental results of different reconstruction targets, including using masked patches and all patches. From these results, it can be seen that only using masked patches to compute loss achieves better classification performance than using all image patches. We infer that using all image patches makes the model focus more on the visible patches, which is effortless to model the locally semantic relationship.

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TABLE VI
TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE INDIAN PINES DATASET

Class	EMP-SVM	SuTabNets	CNN	MSSG	DBDA	FDSSC	SS-Tr	SSFTT	**SS-MTr**	**C-SS-MTr**	**S-SS-MTr**	**SC-SS-MTr**
	96.54±2.07	100.0±0.00	100.0±0.00	98.46±2.55	79.35±12.80	89.04±10.60	99.62±1.15	100.0±0.00	99.62±1.15	100.0±0.00	100.0±0.00	100.0±0.00
	63.74±7.45	73.26±8.21	78.96±6.22	87.42±6.17	86.89±10.74	92.35±6.46	68.00±5.20	85.38±3.92	71.31±4.56	75.02±4.39	85.99±4.45	86.73±3.02
	76.56±4.40	88.94±5.54	87.75±5.96	87.80±6.91	86.88±8.79	87.91±11.64	82.05±4.23	87.63±5.69	82.90±3.68	83.36±4.71	92.09±4.73	92.15±5.24
	81.94±5.14	99.23±0.90	98.53±2.56	96.22±7.07	81.64±11.47	80.76±14.64	96.96±3.15	98.06±2.43	98.02±3.10	96.27±3.89	96.68±4.05	98.20±2.42
	86.57±3.85	92.12±3.24	92.18±3.21	88.70±6.28	97.02±3.74	98.69±1.36	87.21±4.52	94.15±2.09	87.49±4.58	89.05±4.07	93.87±2.48	92.96±3.14
	92.85±4.63	95.10±2.68	95.41±2.53	97.69±1.78	96.54±1.25	97.95±1.56	89.54±6.27	97.98±1.26	91.54±7.34	94.46±6.44	98.48±1.63	98.41±1.86
	92.50±6.12	100.0±0.00	100.0±0.00	100.0±0.00	58.08±28.22	66.78±30.24	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	94.10±3.25	99.78±0.27	99.80±0.59	99.93±0.20	99.62±0.58	99.93±0.10	99.85±0.31	99.34±1.22	99.85±0.46	99.98±0.66	99.98±0.07	99.91±0.26
	98.00±6.00	100.0±0.00	100.0±0.00	100.0±0.00	32.65±11.22	32.03±21.44	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	68.28±7.20	85.84±5.32	88.93±3.10	84.79±8.70	81.08±8.08	79.27±12.02	80.02±6.00	89.66±4.47	80.77±5.91	84.76±4.44	89.55±4.12	89.46±6.22
	59.22±4.69	83.25±7.11	87.41±5.47	86.75±8.83	95.43±4.19	92.20±6.64	75.59±5.37	84.72±5.24	77.75±6.88	82.81±7.53	86.89±6.23	87.88±4.88
	66.61±7.68	80.98±6.97	83.84±6.23	83.58±12.48	90.57±13.45	87.94±14.32	74.31±4.72	77.40±4.99	78.02±7.26	76.47±5.76	85.86±4.88	87.47±6.01
	97.08±1.81	99.40±1.52	99.57±0.89	100.0±0.00	91.42±5.35	92.61±4.98	94.11±4.43	98.86±2.15	99.08±1.14	98.76±1.79	98.76±2.58	99.19±1.28
	86.39±5.68	97.37±2.07	97.39±1.71	99.32±0.96	98.43±1.05	98.57±8.41	93.16±3.39	98.75±1.18	93.69±2.77	95.10±3.40	98.10±1.65	97.73±2.57
	71.48±8.55	97.81±2.66	97.19±2.60	97.32±3.96	88.99±5.46	83.25±12.18	92.35±6.11	96.91±2.02	94.73±3.55	95.19±3.84	98.61±1.30	97.54±1.94
	95.48±4.71	99.54±0.65	99.18±0.91	99.32±0.68	68.09±12.75	80.42±6.41	94.93±4.02	100.0±0.00	95.48±5.38	97.81±2.14	97.95±2.54	98.63±1.50
OA (%)	73.32±2.25	87.58±2.31	89.62±1.72	90.60±2.03	90.26±3.06	89.71±2.72	81.92±2.47	90.10±1.58	83.71±2.56	86.25±2.82	91.55±1.59	91.90±1.83
AA (%)	82.96±1.31	93.29±1.01	94.14±0.72	94.21±1.46	83.29±2.47	84.98±3.97	89.23±1.46	94.30±0.83	90.65±1.55	91.81±1.40	95.18±0.80	95.39±0.95
K×100	69.93±2.50	85.89±2.59	88.20±1.93	89.27±2.30	88.95±3.43	88.30±3.08	79.59±2.74	88.76±1.78	81.59±2.85	84.41±3.15	90.39±1.78	90.78±2.06

TABLE VII
TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA UNIVERSITY DATASET

Class	EMP-SVM	SuTabNets	CNN	MSSG	DBDA	FDSSC	SS-Tr	SSFTT	**SS-MTr**	**C-SS-MTr**	**S-SS-MTr**	**SC-SS-MTr**
	81.27±6.60	86.53±3.16	90.85±4.53	97.06±3.69	98.74±1.20	98.59±1.61	76.43±7.00	86.16±6.15	86.42±7.53	92.85±4.37	96.85±2.09	97.34±2.14
	83.13±3.26	95.44±4.44	93.83±3.49	92.29±5.91	99.51±0.36	99.19±0.38	84.77±4.48	94.50±3.60	84.04±5.68	88.09±4.15	95.18±3.83	95.61±3.73
	81.60±4.51	97.25±2.48	98.12±1.27	99.97±0.10	90.81±12.10	92.84±5.72	88.63±3.30	93.94±4.13	93.66±4.53	96.26±4.71	98.87±0.75	98.83±0.95
	95.29±2.44	98.04±1.13	96.29±1.39	97.12±1.38	92.74±7.92	94.75±5.75	67.14±7.81	88.97±5.25	82.62±6.46	91.79±3.39	96.83±1.84	97.14±2.58
	99.26±0.26	99.84±0.22	99.33±0.52	100.0±0.00	99.53±0.63	99.88±0.12	89.37±7.60	98.95±1.06	97.36±2.81	95.64±5.15	99.05±0.91	98.49±2.04
	80.27±6.31	98.32±1.70	99.47±0.63	99.40±1.58	90.96±5.45	95.65±2.21	97.29±2.48	96.07±3.90	95.83±4.53	97.89±1.97	99.59±0.63	99.45±0.55
	93.11±1.56	99.21±0.59	99.39±0.68	100.0±0.00	93.80±8.83	96.36±2.67	95.35±3.62	99.58±0.67	99.16±0.94	99.31±0.99	99.81±0.48	99.51±0.59
	83.86±3.96	95.82±2.07	98.94±0.80	98.99±1.43	89.83±6.47	83.63±11.84	91.74±7.70	86.63±7.99	92.64±5.04	95.61±3.82	98.04±1.87	98.66±1.16
	99.85±0.12	97.76±1.21	96.66±1.33	99.47±0.93	96.82±1.69	97.35±3.44	67.41±7.45	95.54±2.17	79.36±9.04	83.99±8.82	96.36±6.35	98.88±1.17
OA (%)	84.53±2.22	95.01±2.24	95.26±1.74	95.79±2.55	95.87±1.85	96.07±1.89	84.57±2.78	92.61±1.92	87.68±2.69	91.78±1.86	96.79±1.86	97.16±1.64
AA (%)	88.63±1.57	96.47±0.89	96.99±0.74	98.25±0.83	94.75±2.36	95.36±1.64	84.23±1.62	93.37±1.24	90.12±2.07	93.49±1.76	97.84±1.00	98.21±0.58
K×100	80.00±2.76	93.47±2.87	93.80±2.23	94.54±3.24	94.58±2.41	94.84±2.42	80.15±3.39	90.29±2.47	84.17±3.31	89.36±2.33	95.80±2.40	96.25±2.11

3) Ablation Experiment With the Masking Operation: Masking operation plays an important role in the proposed methods for injecting local inductive bias into the Transformer. Also, Table XI shows the OAs of SS-MTr-based methods with and without masking operation. It can be seen that the masking operation is vital for SS-MTr-based methods. For example, the S-SS-MTr with masking outperforms the one without masking by 4.8 percentage points in terms of OA on the Indian Pines dataset.

E. Parameter Analysis

Some hyperparameters, which have influence on the classification performance to a certain extent, are analyzed in this section. Also, the relative experiments can be summarized into the following three parts.

First, there are three pretraining tasks involved in the proposed SS-MTr-based methods, and different tasks may require a different number of epochs for sufficient pretraining. Therefore, experiments with different numbers of pretraining epochs are conducted for the proposed SS-Tr-based methods.

Table XII shows the OAs obtained by different methods. For SS-MTr, the OA obtained by pretraining with 600 epochs is lower than that obtained by 300 epochs on the Pavia University and Houston datasets, which means overfitting with longer pretraining. However, the OAs are close to each other on the Indian Pines dataset. For S-SS-MTr, it can be seen that

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TABLE VIII
TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE HOUSTON DATASET

Class	EMP-SVM	SuTabNets	CNN	MSSG	DBDA	FDSSC	SS-Tr	SSFTT	**SS-MTr**	**C-SS-MTr**	**S-SS-MTr**	**SC-SS-MTr**
	92.99±4.30	95.38±4.10	92.59±4.56	93.44±4.63	93.48±5.59	96.41±2.30	79.95±6.28	95.36±3.62	80.87±8.67	87.55±4.13	92.25±3.98	93.06±3.78
	93.06±5.72	95.84±2.93	97.00±2.21	95.37±2.36	95.10±3.78	97.64±2.11	75.20±7.35	98.58±1.13	76.22±8.15	89.67±4.87	97.03±2.88	98.21±1.48
	98.97±1.10	98.22±1.17	98.66±1.41	98.67±1.26	100.0±0.00	100.0±0.00	97.15±1.77	99.20±0.60	97.27±2.28	97.89±2.36	99.51±0.49	99.34±0.65
	94.75±2.94	97.16±2.08	97.66±1.81	97.69±1.74	97.13±2.17	95.63±3.79	60.92±8.77	97.82±3.19	69.93±7.61	88.93±5.65	98.28±1.69	98.68±1.73
	96.51±4.52	98.27±3.85	97.47±5.11	98.61±3.22	97.66±2.42	97.60±2.48	97.90±2.04	99.99±0.02	98.72±1.10	97.83±1.89	98.07±5.22	98.13±5.14
	94.72±3.42	97.31±2.56	95.38±3.52	95.08±3.64	97.37±2.23	99.80±0.34	91.97±4.82	98.42±3.86	90.98±5.65	92.56±4.28	95.57±4.33	96.16±3.66
	85.54±4.67	84.77±5.06	90.54±2.41	91.46±3.66	91.93±3.62	92.56±4.64	74.87±6.62	84.32±5.32	83.51±4.61	88.29±3.20	91.82±2.36	93.07±1.93
	69.36±4.90	78.58±5.54	78.48±6.64	78.02±5.81	94.88±3.19	92.68±3.66	78.24±4.24	80.80±5.19	78.70±4.01	78.46±4.82	86.25±3.00	87.12±4.84
	75.81±6.81	85.77±4.65	90.60±3.94	92.26±3.16	88.57±2.27	90.82±3.19	79.04±4.05	78.82±6.79	82.56±4.67	90.28±3.13	91.53±1.12	92.91±2.28
	87.63±4.01	96.10±3.78	96.06±4.12	97.61±2.10	89.76±3.83	89.12±4.70	96.12±2.61	95.10±3.33	94.10±3.66	96.17±3.58	97.03±1.91	98.11±2.02
	85.58±7.72	89.87±3.95	91.52±4.71	93.82±5.63	95.44±2.10	92.70±3.56	90.58±7.82	95.82±3.23	90.26±5.23	90.56±5.83	93.45±5.76	93.48±6.87
	76.18±6.14	92.00±6.74	91.81±5.48	92.45±6.26	93.15±3.27	93.40±3.86	84.73±5.04	89.65±4.87	84.18±5.41	84.20±9.59	89.79±6.66	88.76±5.20
	56.44±5.90	98.00±2.34	96.08±2.62	95.86±2.48	82.75±6.06	83.22±9.14	92.98±5.95	97.75±2.10	93.16±4.35	91.54±3.83	94.10±2.98	95.32±2.75
	97.94±2.53	99.78±0.44	99.93±0.22	99.95±0.15	98.13±2.59	98.07±2.72	100.0±0.00	99.85±0.22	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
	99.08±0.46	99.86±0.30	99.59±1.06	99.45±1.59	95.05±2.47	96.22±3.18	96.84±2.96	100.0±0.00	98.30±1.59	98.36±1.46	99.77±0.47	99.83±0.32
OA (%)	86.56±1.36	92.60±0.66	93.36±0.92	93.92±1.02	93.67±0.92	93.97±0.99	84.13±1.72	92.88±0.97	85.99±1.87	90.43±1.27	94.33±1.03	94.88±1.24
AA (%)	86.97±1.26	93.79±0.58	94.23±0.67	94.65±0.85	94.03±0.87	94.39±1.00	86.43±1.54	94.10±0.85	87.92±1.72	91.49±0.96	94.96±0.74	95.48±0.99
K×100	85.47±1.47	92.01±0.71	92.82±0.99	93.43±1.10	93.16±1.00	93.48±1.07	82.86±1.86	92.30±1.05	84.87±2.02	89.65±1.38	93.87±1.12	94.46±1.34

TABLE IX
ABLATION EXPERIMENT FOR THE SIA

Datasets	SIA	SS-Tr	SS-MTr	C-SS-MTr	S-SS-MTr	SC-SS-MTr
Indian Pines	w/	81.92	83.71	86.25	91.55	91.90
	w/o	26.18	50.58	55.78	69.83	70.56
Pavia University	w/	84.57	87.68	91.78	96.79	97.15
	w/o	70.61	82.98	83.65	84.86	85.45
Houston	w/	84.13	85.99	90.43	94.33	94.88
	w/o	58.41	81.06	84.69	88.47	89.25

w/: with, w/o: without

TABLE XI
ABLATION EXPERIMENT FOR THE MASKING OPERATION

Datasets	Masking	SS-MTr	C-SS-MTr	S-SS-MTr	SC-SS-MTr
Indian Pines	w/	83.71	86.25	91.55	91.90
	w/o	82.53	84.12	86.75	87.86
Pavia University	w/	87.68	91.78	96.79	97.15
	w/o	86.25	88.93	94.36	95.28
Houston	w/	85.99	90.43	94.33	94.88
	w/o	84.86	88.25	92.67	93.49

w/: with, w/o: without

TABLE X
ABLATION EXPERIMENT FOR THE RECONSTRUCTION TARGET

Datasets	Target	SS-MTr	C-SS-MTr	S-SS-MTr	SC-SS-MTr
Indian Pines	masked	83.71	86.25	91.55	91.90
	all	82.85	85.40	90.92	91.48
Pavia University	masked	87.68	91.78	96.79	97.15
	all	86.86	89.71	96.15	96.63
Houston	masked	85.99	90.43	94.33	94.88
	all	85.56	90.16	93.92	94.40

TABLE XII
PARAMETER ANALYSIS FOR THE NUMBER OF PRETRAINING EPOCHS

Datasets	Epochs	SS-MTr	C-SS-MTr	S-SS-MTr	S-CSS-MTr
Indian Pines	300	83.71	84.94	91.55	91.32
	600	83.94	86.25	91.43	91.90
Pavia University	300	87.68	90.61	96.79	96.65
	600	84.13	91.78	96.85	97.15
Houston	300	85.99	89.48	94.33	94.41
	600	83.04	90.43	94.23	94.88

the OAs obtained by pretraining with 600 epochs are close to the ones obtained by 300 epochs, while for C-SS-MTr and SC-SS-MTr, it can be seen that the OAs obtained by pretraining with 600 epochs are higher than the ones obtained by 300 epochs obviously. It seems that the CL task needs more epochs for pretraining, while 300 epochs are enough for the reconstruction and classification tasks. Therefore, considering the classification accuracy and pretraining time, C-SS-MTr and SC-SS-MTr are pretrained with 600 epochs, while SS-MTr and S-SS-MTr are pretrained with 300 epochs in the experiments.

Second, the hyperparameters related to the model setting, including the value of minimum augmentation scale S, the masking ratio p, the decoder's depth L_d, and width D', are analyzed in the following. Considering accuracies and the number of training epochs in the previous experiments, S-SS-MTr is used in this part for efficiency.

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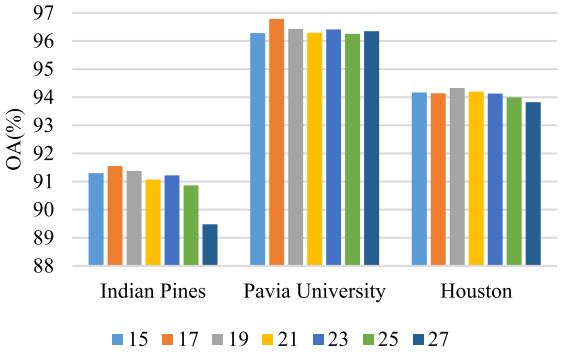


Fig. 9. OA of S-SS-MTr with different values of scale S.

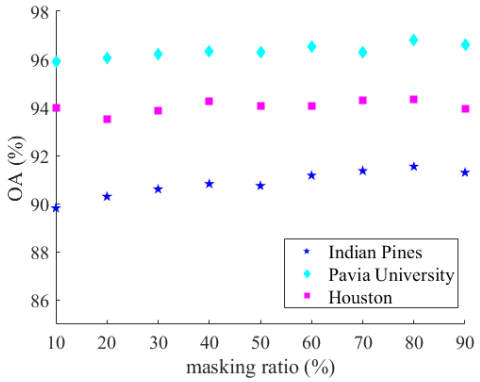


Fig. 10. OA of S-SS-MTr with different values of masking ratio.

1) Minimum Augmentation Scale S: The value of minimum augmentation scale S has an impact over the proposed SS-MTr-based methods. Also, Fig. 9 shows the obtained OAs. Generally speaking, it can be seen that a smaller value of S (e.g., 17 or 19) is preferred over all the three datasets. Specifically, the best values of S for the Indian Pines dataset are 17 and it is the same with the Pavia University dataset. For the Houston dataset, the best classification performance is obtained when $S = 19$. Also, the values of S are set to be optimal for the three datasets in the following experiments.

2) Masking Ratio p: Fig. 10 shows the influence of the masking ratio. It can be seen that the optimal values for masking ratio are surprisingly high. The value of 80% is good for all the three datasets. Besides, the obtained OAs are less sensitive to the values of masking ratio, and a wide range of values (40%–80%) can work well. It is worth noting that a higher value of masking ratio leads to less computational burden. In the experiments, the value of masking ratio is set to be 80%.

3) Decoder's Depth L_d and Width D': As shown in Tables XI and XII, the decoder used in the proposed SS-MTr-based methods can be flexibly designed. Here, S-SS-MTr is used to explore the influence caused by the decoder's depth L_d and width D', just like before.

Table XIII shows the classification accuracies varying with the decoder's depth. It can be seen that the decoder depth is less influential for improving the classification performance. A shallower decoder can also achieve good results. Even when the value of depth is 0, i.e., only a linear layer is used, the obtained accuracies are slightly lower than the ones obtained by other values of depth. For all the three datasets, the best classification performance is obtained when $L_d = 3$.

TABLE XIII
PARAMETER ANALYSIS FOR THE DECODER'S DEPTH

		0	1	2	3	4
Indian Pines	OA (%)	91.07	90.76	91.41	91.55	91.24
	AA (%)	94.98	94.70	95.11	95.18	95.02
	K×100	89.86	89.51	90.27	90.39	90.04
Pavia University	OA (%)	96.04	96.36	96.62	96.79	96.34
	AA (%)	97.67	97.65	98.04	97.84	97.93
	K×100	94.84	95.25	95.58	95.8	95.23
Houston	OA (%)	93.92	94.25	94.09	94.33	94.17
	AA (%)	94.6	94.97	94.79	94.96	94.84
	K×100	93.43	93.79	93.61	93.87	93.69

TABLE XIV
PARAMETER ANALYSIS FOR THE DECODER'S WIDTH

		32	64	128	256	512
Indian Pines	OA (%)	91.37	91.47	91.55	91.34	91.26
	AA (%)	95.11	95.05	95.18	95.04	95.12
	K×100	90.20	90.31	90.39	90.15	90.06
Pavia University	OA (%)	96.51	96.48	96.79	96.40	96.68
	AA (%)	97.99	97.78	97.84	97.94	97.89
	K×100	95.43	95.41	95.80	95.30	95.66
Houston	OA (%)	94.18	94.42	94.33	94.20	94.12
	AA (%)	94.89	95.09	94.96	94.88	95.08
	K×100	93.71	94.06	93.87	93.73	93.75

Table XIV shows the classification accuracies varying with the decoder's width. It can be seen that the decoder width has little influence on the classification results for all the three datasets. For example, the differences between OAs obtained with $D' = 64$ and $D' = 128$ on the three datasets are -0.0008, -0.0031, and 0.0009.

Overall, the decoder for S-SS-MTr can be lightweight, which will lead to less computational burden. For simplicity, the values of depth L_d and width D' are set to be 3 and 128, respectively.

Finally, the weight coefficient of classification loss λ and the weight coefficient of CL loss γ are analyzed. For efficiency, we first analyze the effect of different values of λ using S-SS-MTr. Then, the value of λ is fixed and different values of γ are analyzed for SC-SS-MTr.

a) Weight coefficient λ: For S-SS-MTr, the influence of different values of classification loss coefficient is analyzed in this part. Fig. 11 shows the OAs varying with different values of λ on the three datasets. It can be seen that too large or too small values of λ leads to bad classification performance. A too large value of λ forces S-SS-MTr to degrade into ordinary supervised pretraining, which is harmful for performance, while a too large value of λ forces S-SS-MTr to degrade into SS-MTr, which is also harmful for the classification performance. Specifically, the optimal values are 0.005, 0.005, and 0.01 for the Indian Pines dataset, the Pavia University dataset, and the Houston dataset, respectively.

b) Weight coefficient γ: In this part, SC-SS-MTr is used for exploring the influence of the value of CL loss coefficient γ

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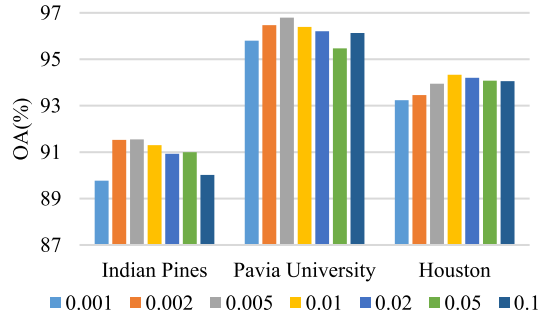


Fig. 11. OA of S-SS-MTr with different values of classification loss ratio λ.

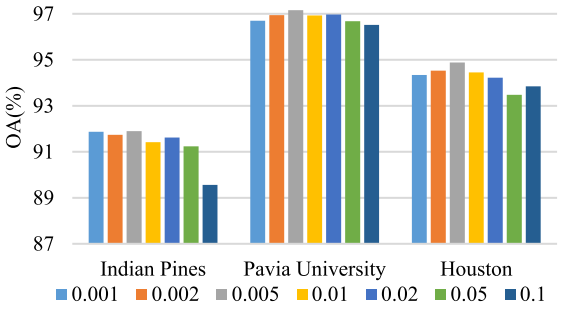


Fig. 12. TOA of SC-SS-MTr with different values of CL loss ratio γ.

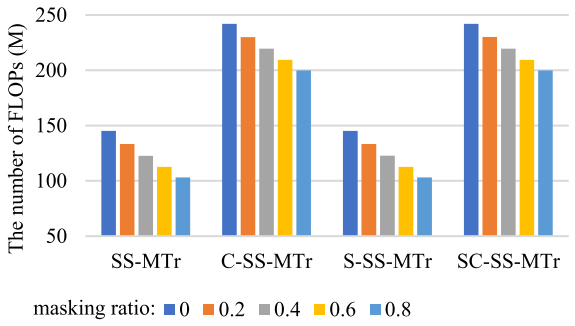


Fig. 13. Number of FLOPs of SS-MTr-based methods with various masking ratios.

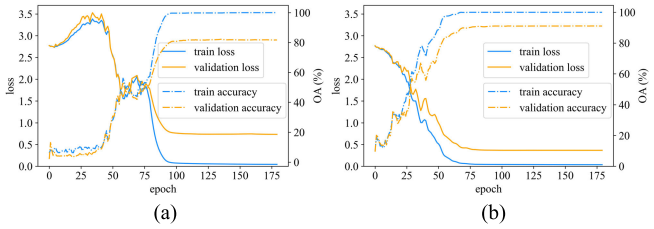


Fig. 14. Training procedure of SS-Tr on the Indian Pines dataset: (a) without S-SS-MTr pretraining and (b) with S-SS-MTr pretraining.

after the optimal values of λ have been found in the previous experiments. Fig. 12 shows the OAs varying with different values of γ on the three datasets. Like the analysis of λ in the previous experiments, the values of γ should be set not to be too large or too small. The optimal values for all the three datasets are 0.005.

Note that we set the values of γ for C-SS-MTr the same as SC-SS-MTr on the three datasets.

F. Algorithm Complexity

Table XV shows the algorithm complexity for different classification methods (take Pavia University dataset as an example). FLOPs, which means the number of floating-point

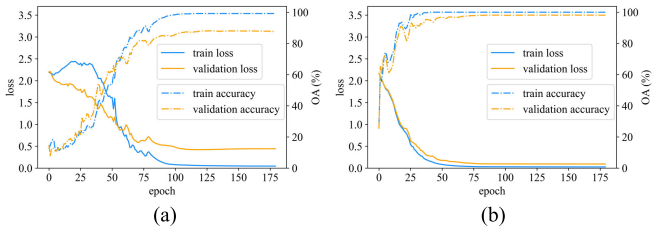


Fig. 15. Training procedure of SS-Tr on the Pavia University dataset: (a) without S-SS-MTr pretraining and (b) with S-SS-MTr pretraining.

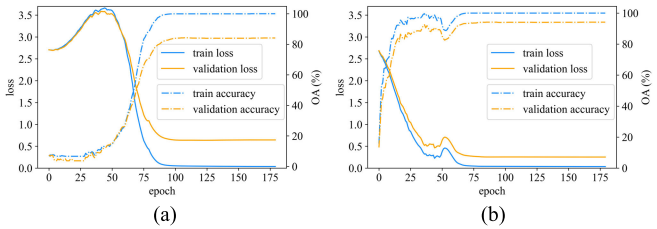


Fig. 16. Training procedure of SS-Tr on the Houston dataset: (a) without S-SS-MTr pretraining and (b) with S-SS-MTr pretraining.

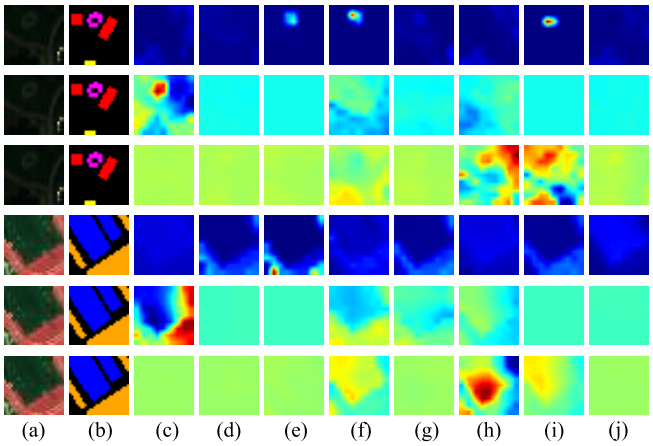


Fig. 17. Attention maps using different methods on the Houston dataset. (a) Input HSI image (false color), (b) ground truth, and (c)–(j) 1st–8th heads' attention maps for the encoder's last block. The first and fourth rows show the maps of the SS-Tr, the second and fifth rows show maps of the S-SS-MTr after pretraining, and the third and last rows show maps of the S-SS-MTr after fine-tuning.

TABLE XV
ANALYSIS OF FLOPS AND PARAMETERS

	MSSG	SS-Tr	SSFTT	SS-MTr	C-SS-MTr	S-SS-MTr	SC-SS-MTr
FLOPs	93.44G	84.22M	11.71M	103.06M	199.92M	103.09M	199.95M
Param.	0.1M	0.7M	0.2M	1.3M	1.5M	1.3M	1.5M

operations needed for a given model, is used here to measure the amount of calculation in the feedforward propagation. Param. means the number of parameters to be trained in a given model. The FLOPs and Param. can be used to measure the complexity of a given deep model.

The proposed SS-MTr-based methods are pretraining frameworks, which use SS-Tr as backbone. Therefore, their numbers of FLOPs and parameters are related to the SS-Tr. Compared with SS-Tr, SS-MTr has two additional parts in structure: 1) SS-MTr uses masking operation, which helps reduce the

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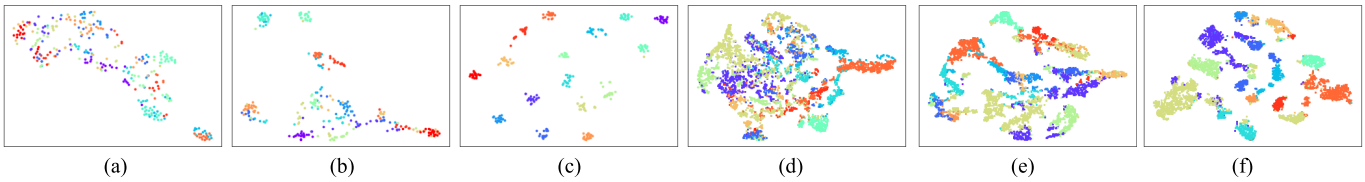


Fig. 18. Visualization of pretraining features using t-SNE on the Indian Pines dataset: (a) SS-MTr over training set, (b) C-SS-MTr over training set, (c) S-SS-MTr over training set, (d) SS-MTr over validation set, (e) C-SS-MTr over validation set, and (f) S-SS-MTr over validation set.

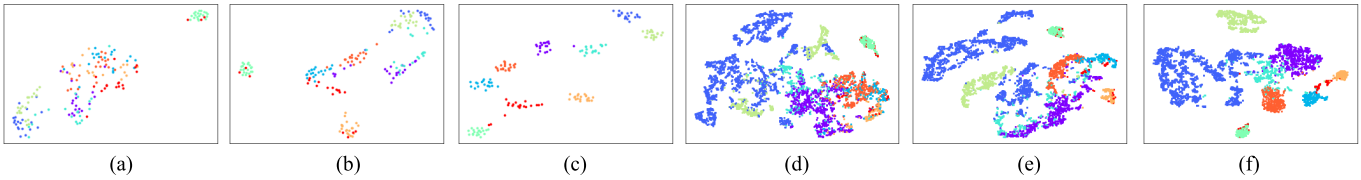


Fig. 19. Visualization of pretraining features using t-SNE on the Pavia University dataset: (a) SS-MTr over training set, (b) C-SS-MTr over training set, (c) S-SS-MTr over training set, (d) SS-MTr over validation set, (e) C-SS-MTr over validation set, and (f) S-SS-MTr over validation set.

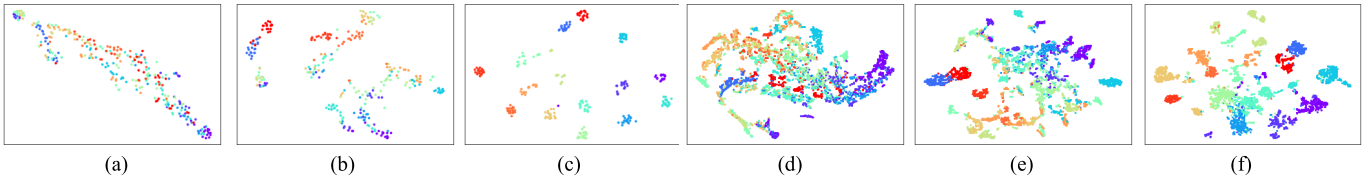


Fig. 20. Visualization of pretraining features using t-SNE on the Houston dataset: (a) SS-MTr over training set, (b) C-SS-MTr over training set, (c) S-SS-MTr over training set, (d) SS-MTr over validation set, (e) C-SS-MTr over validation set, and (f) S-SS-MTr over validation set.

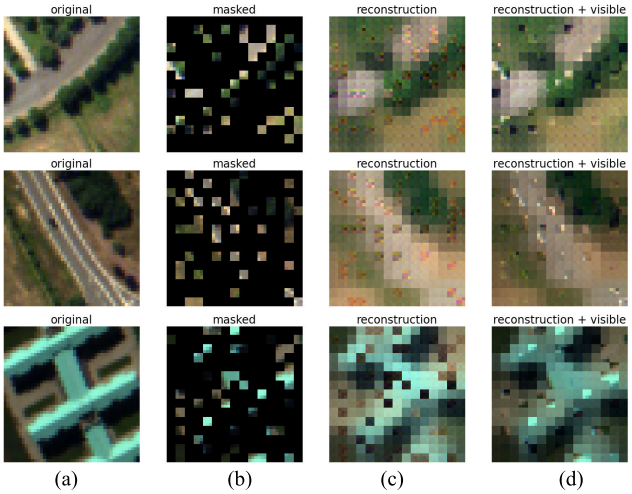


Fig. 21. Reconstruction examples obtained by S-SS-MTr with 80% masking ratio on the Pavia University dataset: (a) original images, (b) masked images, (c) reconstruction images, and (d) reconstruction and visible images.

computational effort, and 2) SS-MTr uses a decoder consisting of several (e.g., three in this study) Transformer blocks, which leads to an increase in the computational effort and parameters. The computational effort increased by using the decoder is greater than that reduced by the masking operation, and thus, the total computational effort increases. In fact, when the masking ratio is 0, the calculation amount of SS-MTr in Fig. 13 is 145.16M, whereas the calculation amount of SS-Tr is 84.22M in Table XV. Due to the masking operation, the computational burden of SS-MTr does not increase much when compared with SS-Tr, which can be seen in Table XV. Due to the masking operation, the computational burden of SS-MTr does not increase much when compared with SS-Tr. Compared with SS-MTr, the adding of CL branch increases the number of parameters by 0.3M and FLOPs by 96.3M. However, the supervised learning branch brings less algorithm complexity according to FLOPs and parameters. Meanwhile, as a global learning method, MSSG takes the whole HSI scene as input and thus requires more computational resources in the training process.

Fig. 13 shows the numbers of FLOPs for SS-MTr-based methods with different masking ratios. It can be seen that a larger value of masking ratio means fewer FLOPs. It is worth noting that the masking operation cannot change the number of parameters because of the special mechanism of the Transformer.

G. Visualization

In this section, we conduct five visualization experiments to help understand the proposed methods. To save time and space, not all SS-MTr-based methods are evaluated in this section.

1) Visualization for the Training Process With/Without Pretraining: In order to further demonstrate the effectiveness of the pretraining process, we plot the training convergence curves of SS-Tr with/without S-SS-MTr's pretraining. Figs. 14–16 show the train loss, train accuracy, validation loss, and validation accuracy in the training procedure. It can be seen that the pretraining benefits the SS-Tr's training significantly. The train accuracies obtained by SS-Tr with S-SS-MTr's pretraining and without pretraining are both 100% in the later training process, while the validation accuracies obtained by SS-Tr with pretraining are higher than the ones

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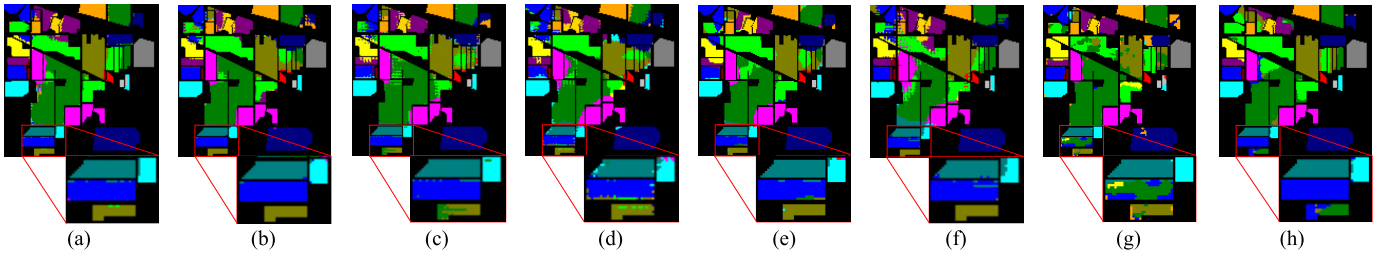


Fig. 22. Classification maps using different methods on the Indian Pines dataset. (a) SC-SS-MTr. (b) S-SS-MTr. (c) C-SS-MTr. (d) SS-MTr. (e) SSFTT. (f) SS-Tr. (g) FDSCC. (h) MSSG.

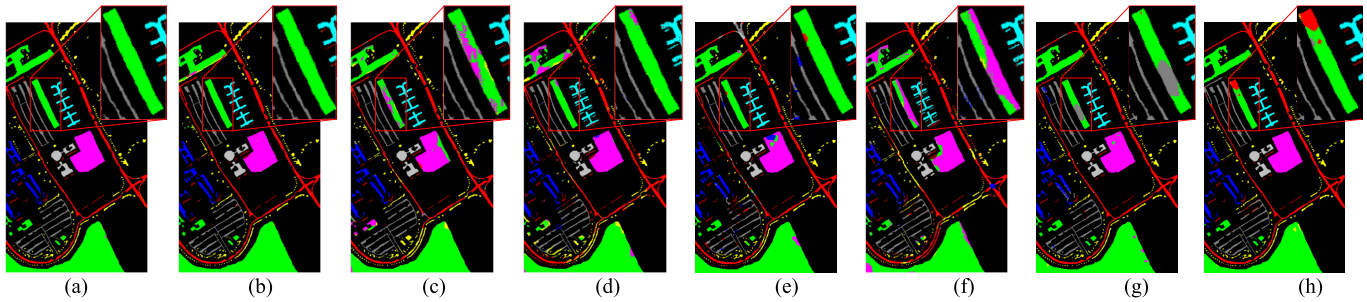


Fig. 23. Classification maps using different methods on the Pavia University dataset. (a) SC-SS-MTr. (b) S-SS-MTr. (c) C-SS-MTr. (d) SS-MTr. (e) SSFTT. (f) SS-Tr. (g) FDSCC. (h) MSSG.

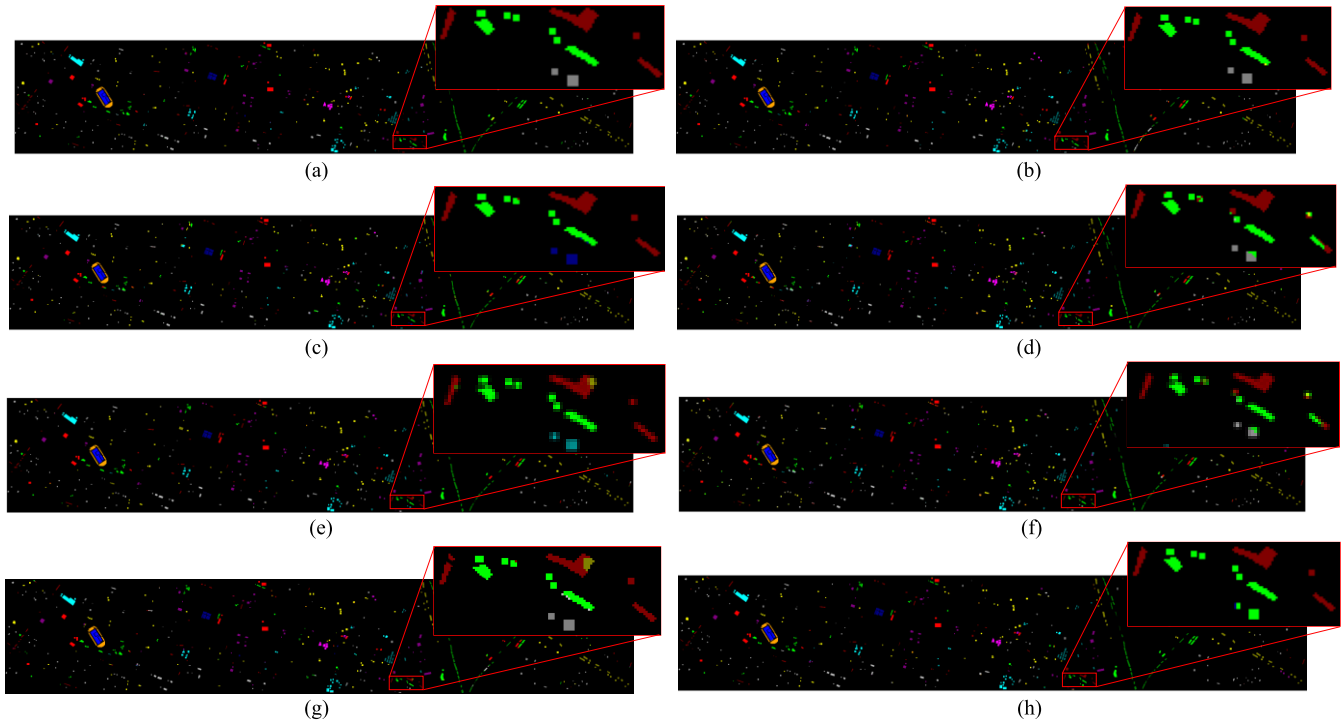


Fig. 24. Classification maps using different methods on the Houston dataset. (a) SC-SS-MTr. (b) S-SS-MTr. (c) C-SS-MTr. (d) SS-MTr. (e) SSFTT. (f) SS-Tr. (g) FDSCC. (h) MSSG.

without pretraining. It indicates that the S-SS-MTr pretraining helps SS-Tr alleviate the overfitting problem. Besides, the SS-Tr has faster and better convergence with the help of pretraining. The similar phenomena can be found in the other proposed methods.

2) Visualization for Attention Maps: Fig. 17 shows the attention maps obtained by different models on the Houston dataset. From these results, it can be seen that the attention weights obtained by SS-Tr are much sparser than the ones obtained after S-SS-MTr's pretraining and fine-tuning, which means that the SS-Tr is prone to be overfitting. Besides, SS-Tr focuses less on the image center, which contradicts the fact that the image label coincides with the label of the center pixel. However, it can be seen that the pretraining improves the attention results and helps Transformer generalize better.

3) Visualization for Features With Different Models: Figs. 18–20 show the pretraining features' t-SNE distributions for SS-MTr, C-SS-MTr, and S-SS-MTr on the three datasets. From these results, it can be seen that the features learned by SS-MTr pretraining are less discriminative. With the help of

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CL, the features learned by C-SS-MTr have better separability with regard to the training set and validation set. Also, S-SS-MTr can learn the most discriminative features among these models.

4) Visualization for Reconstruction Results: Fig. 21 shows the reconstruction HSI images for the Pavia University dataset. For clarity, we generate the input images with a spatial size of 45×45. These results show that the obtained images reconstructed by S-SS-MTr are similar to the original HSI images to some extent. Specifically, the detailed information is lost in the reconstruction images, while roughly contextual information has been reserved. It is worth noting that the reconstruction results are similar to other SS-MTr-based methods.

5) Classification Maps for Different Models: Figs. 22–24 show the classification maps for different methods on the three datasets. The classification performance obtained by the proposed methods is better than other methods, which can be seen in the classification maps clearly.

V. Conclusion

In this study, an SS-MTr is proposed for HSI classification, which focuses on how to train a Transformer effectively, especially with limited training samples. Furthermore, discriminative feature learning is incorporated with SS-MTr for better classification performance, leading to three proposed D-SS-MTr methods (i.e., C-SS-MTr, S-SS-MTr, and SC-SS-MTr). The experiments are conducted for verification on the three widely used datasets (i.e., Indian Pines, Pavia University, and Houston datasets), and some main conclusions can be drawn from the results as follows.

1) According to the comparative analysis of classification results, the proposed SS-MTr-based methods help ordinary Transformer gain better classification performance. In particular, the proposed SC-SS-MTr outperforms some existing state-of-the-art HSI classification methods in terms of OA, AA, and K.
2) CL can be combined with SS-MTr for adding discriminative information and improves the downstream HSI classification performance, that is, the CL task in the C-SS-MTr is effective.
3) The utilization of supervision labels can help SS-MTr leverage global and discriminative information well and S-SS-MTr gain competitive classification results.
4) The combination of CL task, supervised classification task, and the reconstruction task can work together to pretrain a Transformer for HSI classification with limited training samples.

However, the proposed methods utilized a two-stage learning strategy to train a Transformer-based backbone, which leads to heavy training burden and inconvenience when used for practice. Besides, the proposed methods have the potential to utilize unlabeled samples that are abundant in HSI. Therefore, the future work will improve the proposed methods to be end-to-end and lightweight, and we will extend them to unsupervised and semisupervised settings.

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